Structured Encoder for Ancient Logograms

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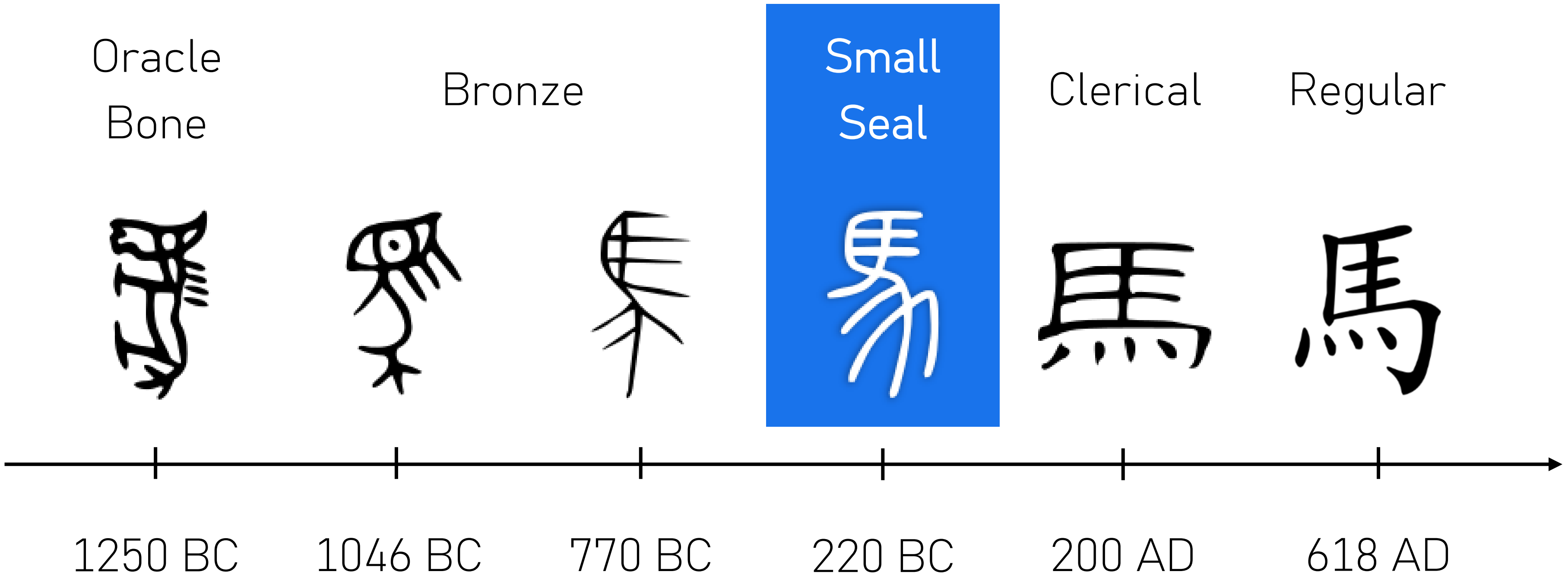
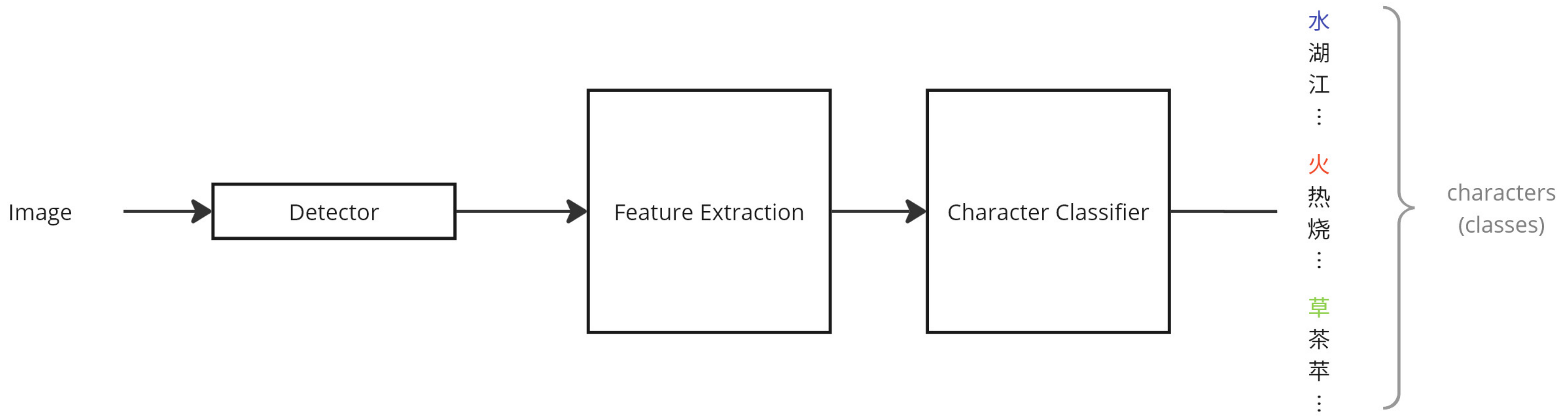
Evolution of Chinese Scripts



Straightforward Approach

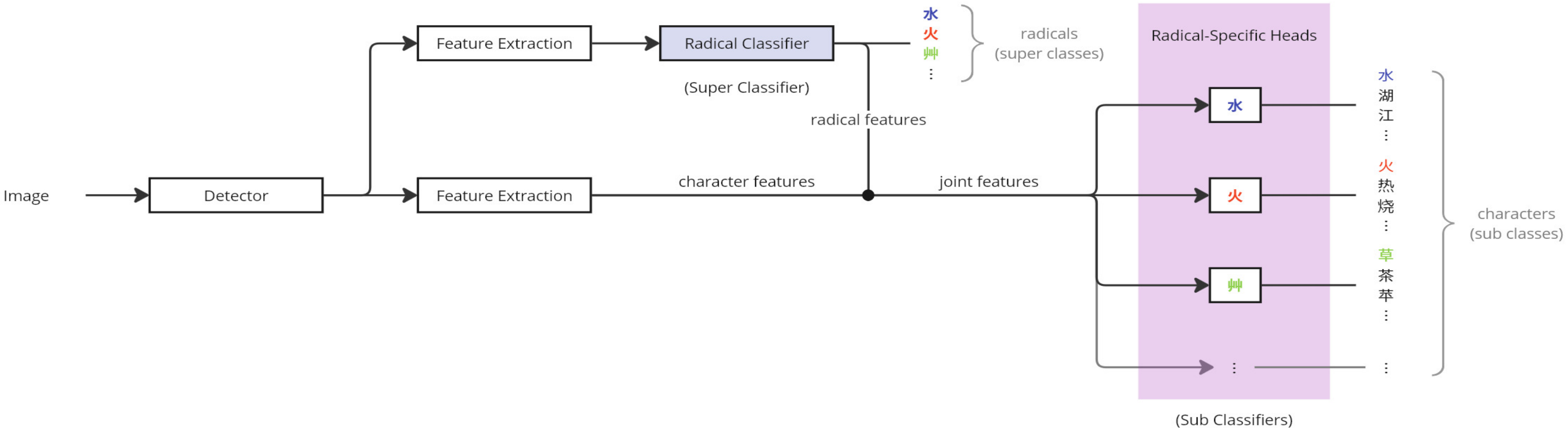


Radicals Examples

Radical	water		fire	
Script	Regular	Seal	Regular	Seal
	水	水	火	火

Characters	lake	river	hot	burn
Regular Script	湖	江	熱	燒
Seal Script	湖	江	熱	燒

Hierarchical Approach



Dataset

SEAL 5684

Characters selection

Kangxi Dictionary
(1716 AD)

40k entries

∩

UNICODE

CJK Unified
Ideographs

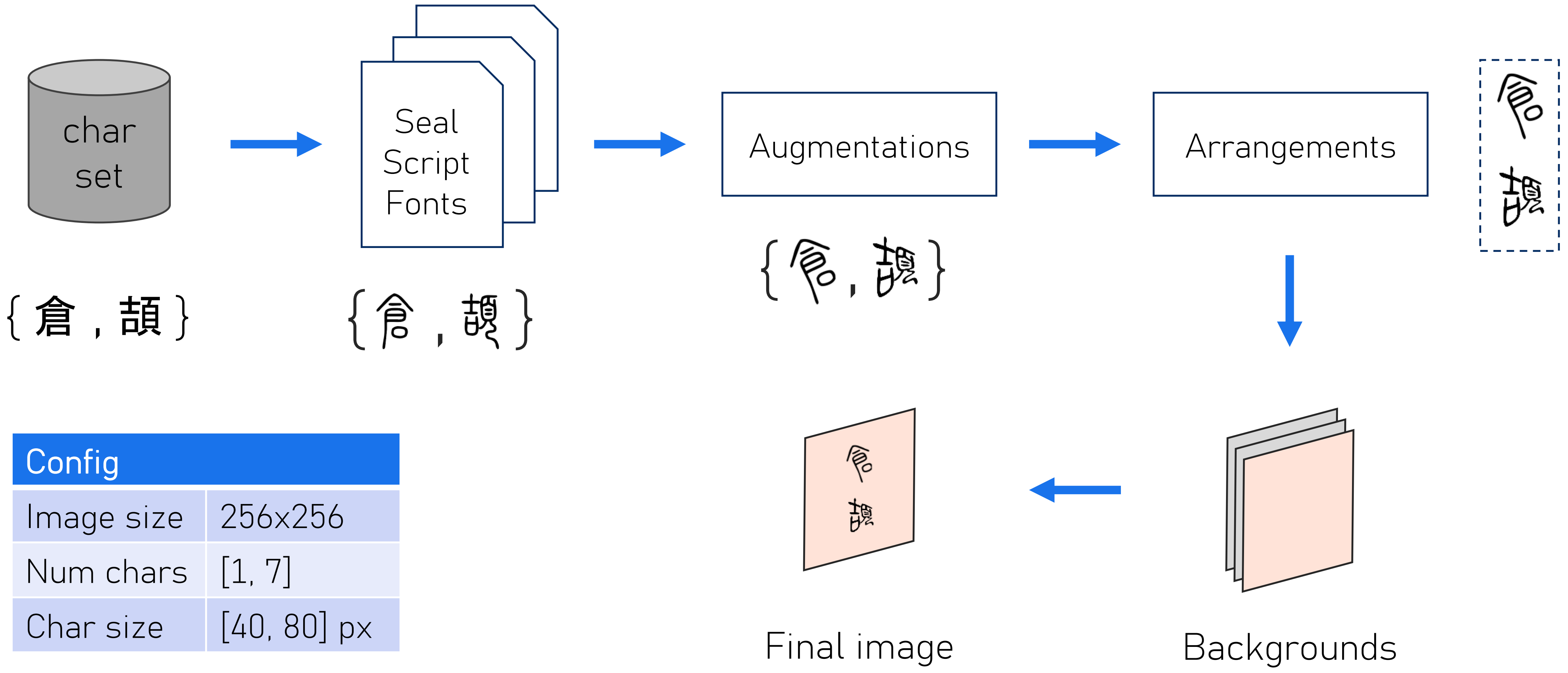
=

16k Renderable
Classical Characters

Seal
Script
Fonts

5684 characters set

Image generation



Config	
Image size	256x256
Num chars	[1, 7]
Char size	[40, 80] px

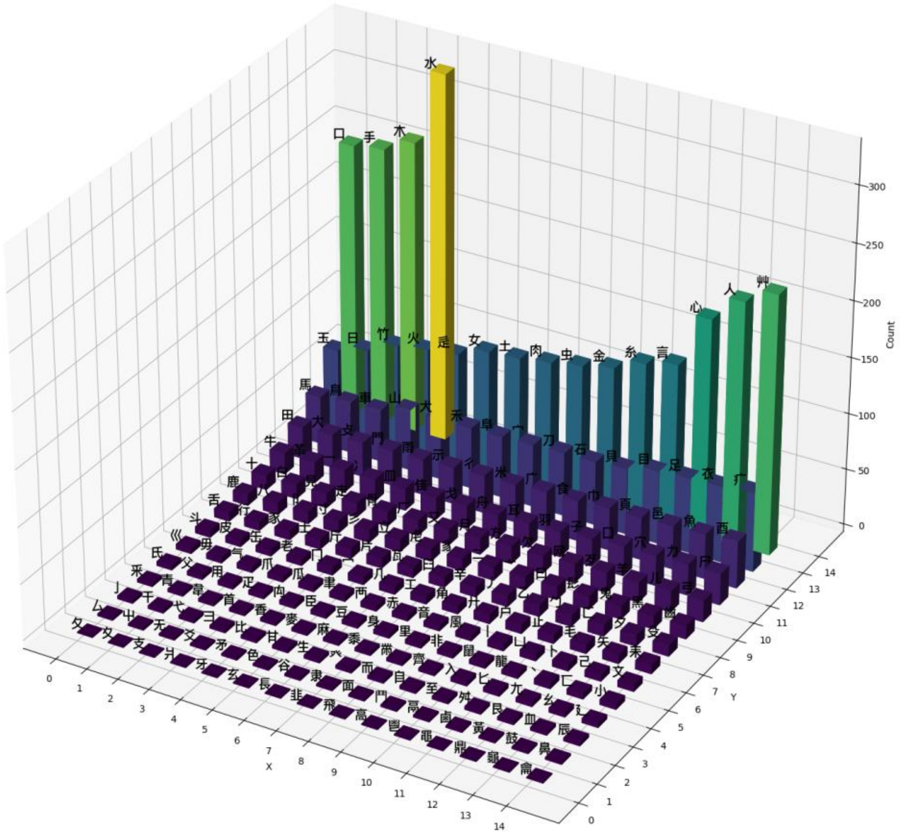
Splits

- How many images? → Coupon collector's problem, rounded up
- 90% Train (35k) / 10% test (4k)
- Labels format:
  - SEAL: `char x1 y1 x2 y2 radical`
  - YOLO: `char_id cx cy bw bh`
- Calculated image mean and std over the train split

Class weights

- High (intrinsic) imbalance in radicals' distribution
- Need superclass weights (weighted CE)
- Also did for subclasses (although more uniform)

$$w_c = \frac{\sum_{i \in C} n_i}{|C| n_c}$$

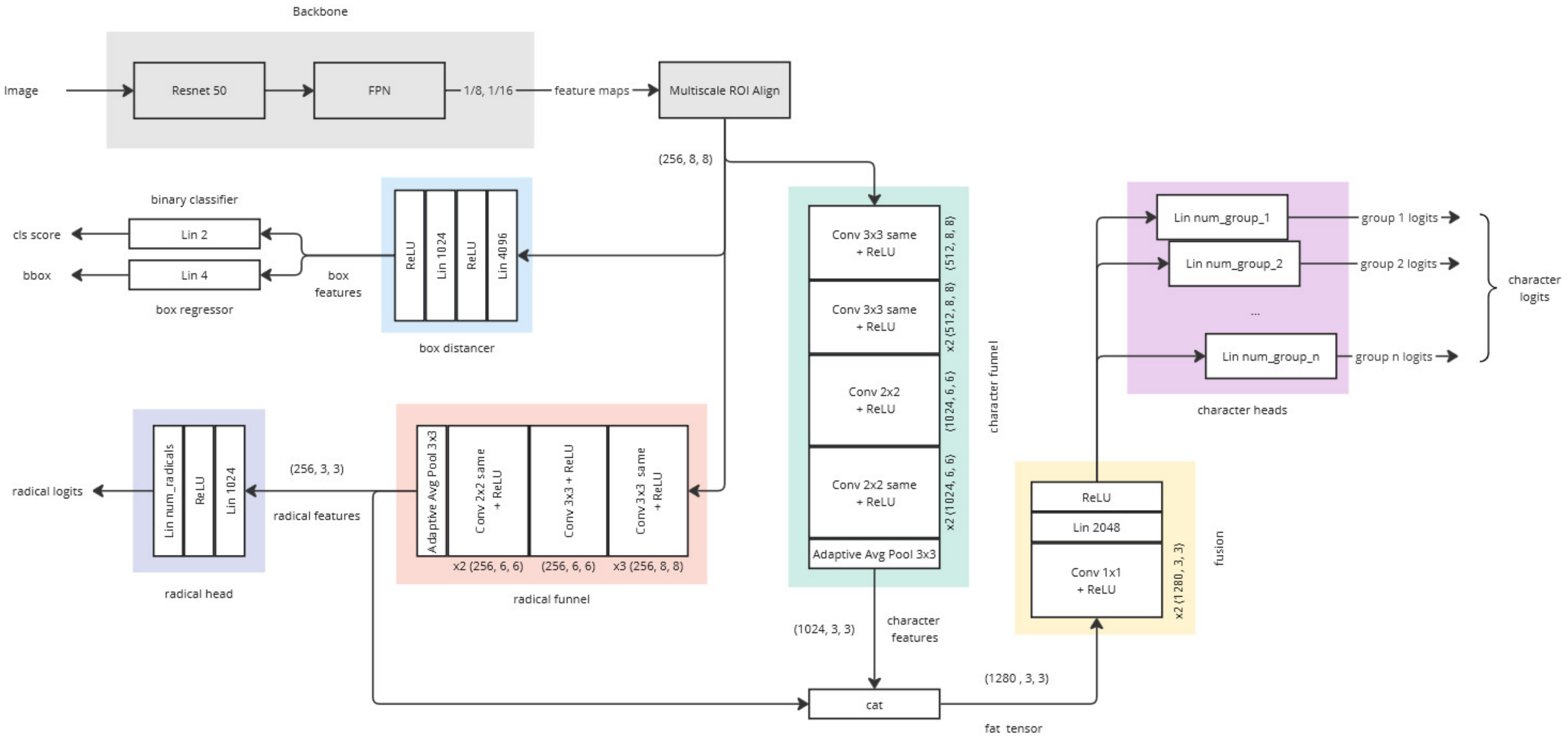


Model

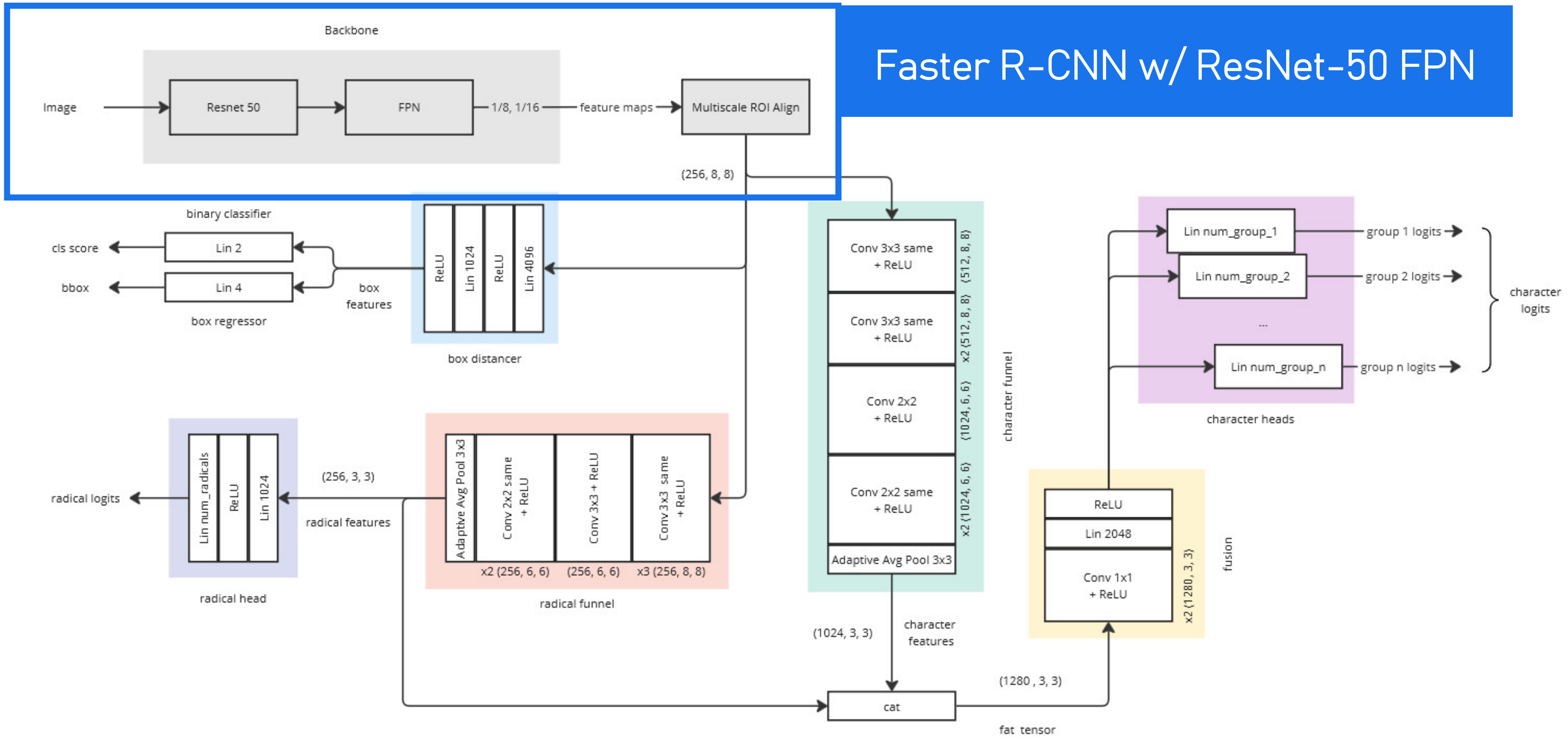
Building SEAL

```
(cls_score_dummy): Linear(in_features=1024, out_features=2, bias=True)
(bbox_pred): Linear(in_features=1024, out_features=8, bias=True)
(superclassifier_funnel): Sequential(
  (0): Conv2d(256, 256, kernel_size=(3, 3), stride=(1, 1), padding=(1, 1))
  (1): ReLU()
  (2): Conv2d(256, 256, kernel_size=(3, 3), stride=(1, 1), padding=(1, 1))
  (3): ReLU()
  (4): Conv2d(256, 256, kernel_size=(3, 3), stride=(1, 1), padding=(1, 1))
  (5): ReLU()
  (6): Conv2d(256, 256, kernel_size=(3, 3), stride=(1, 1))
  (7): ReLU()
  (8): Conv2d(256, 256, kernel_size=(2, 2), stride=(1, 1), padding=(1, 1))
  (9): ReLU()
  (10): Conv2d(256, 256, kernel_size=(2, 2), stride=(1, 1), padding=(1, 1))
  (11): ReLU()
  (12): AdaptiveAvgPool2d(output_size=3)
)
(superclassifier_head): Sequential(
  (0): Flatten(start_dim=1, end_dim=-1)
  (1): Linear(in_features=2304, out_features=1024, bias=True)
  (2): ReLU()
  (3): Linear(in_features=1024, out_features=214, bias=True)
)
(subclassifier_funnel): Sequential(
  (0): Conv2d(256, 512, kernel_size=(3, 3), stride=(1, 1), padding=(1, 1))
  (1): ReLU()
  (2): Conv2d(512, 512, kernel_size=(3, 3), stride=(1, 1), padding=(1, 1))
  (3): ReLU()
  (4): Conv2d(512, 512, kernel_size=(3, 3), stride=(1, 1), padding=(1, 1))
  (5): ReLU()
  (6): Conv2d(512, 1024, kernel_size=(3, 3), stride=(1, 1))
  (7): ReLU()
  (8): Conv2d(1024, 1024, kernel_size=(2, 2), stride=(1, 1), padding=(1, 1))
  (9): ReLU()
  (10): Conv2d(1024, 1024, kernel_size=(2, 2), stride=(1, 1), padding=(1, 1))
  (11): ReLU()
  (12): AdaptiveAvgPool2d(output_size=3)
)
(subclassifier_junction): Sequential(
  (0): Conv2d(1280, 1280, kernel_size=(1, 1), stride=(1, 1))
  (1): ReLU()
  (2): Conv2d(1280, 1280, kernel_size=(1, 1), stride=(1, 1))
  (3): ReLU()
  (4): Flatten(start_dim=1, end_dim=-1)
  (5): Linear(in_features=11520, out_features=2048, bias=True)
  (6): ReLU()
)
(sub_classifiers): ModuleList(
  (0): Linear(in_features=2048, out_features=328, bias=True)
```

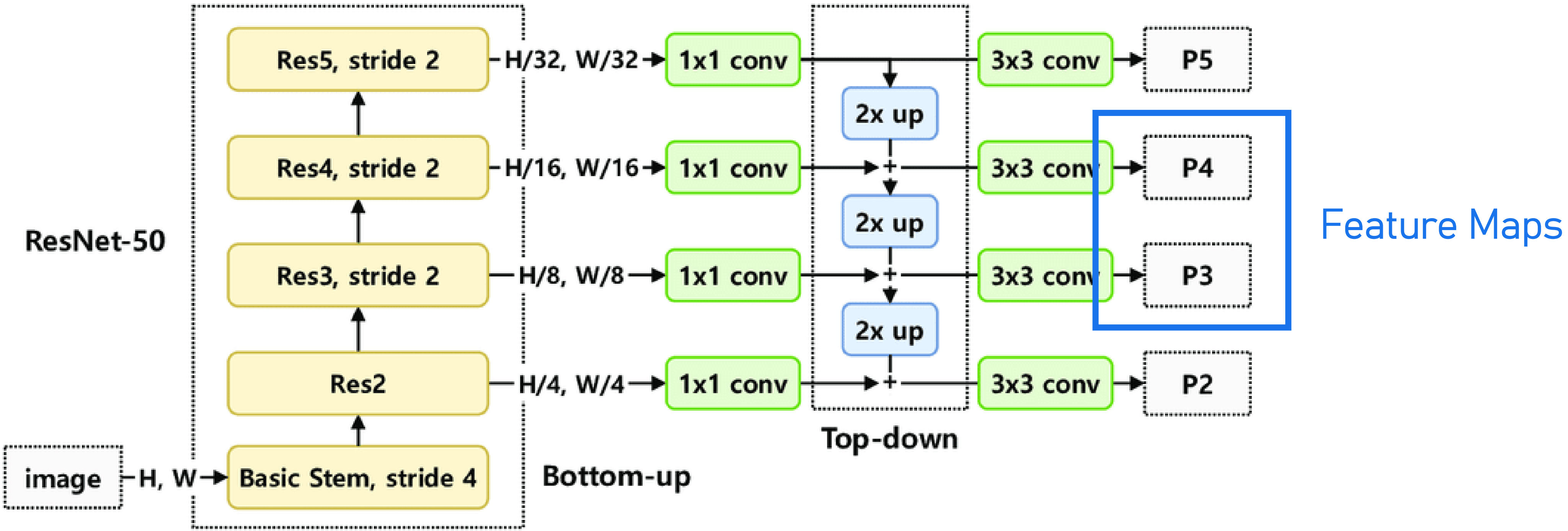
Final Architecture



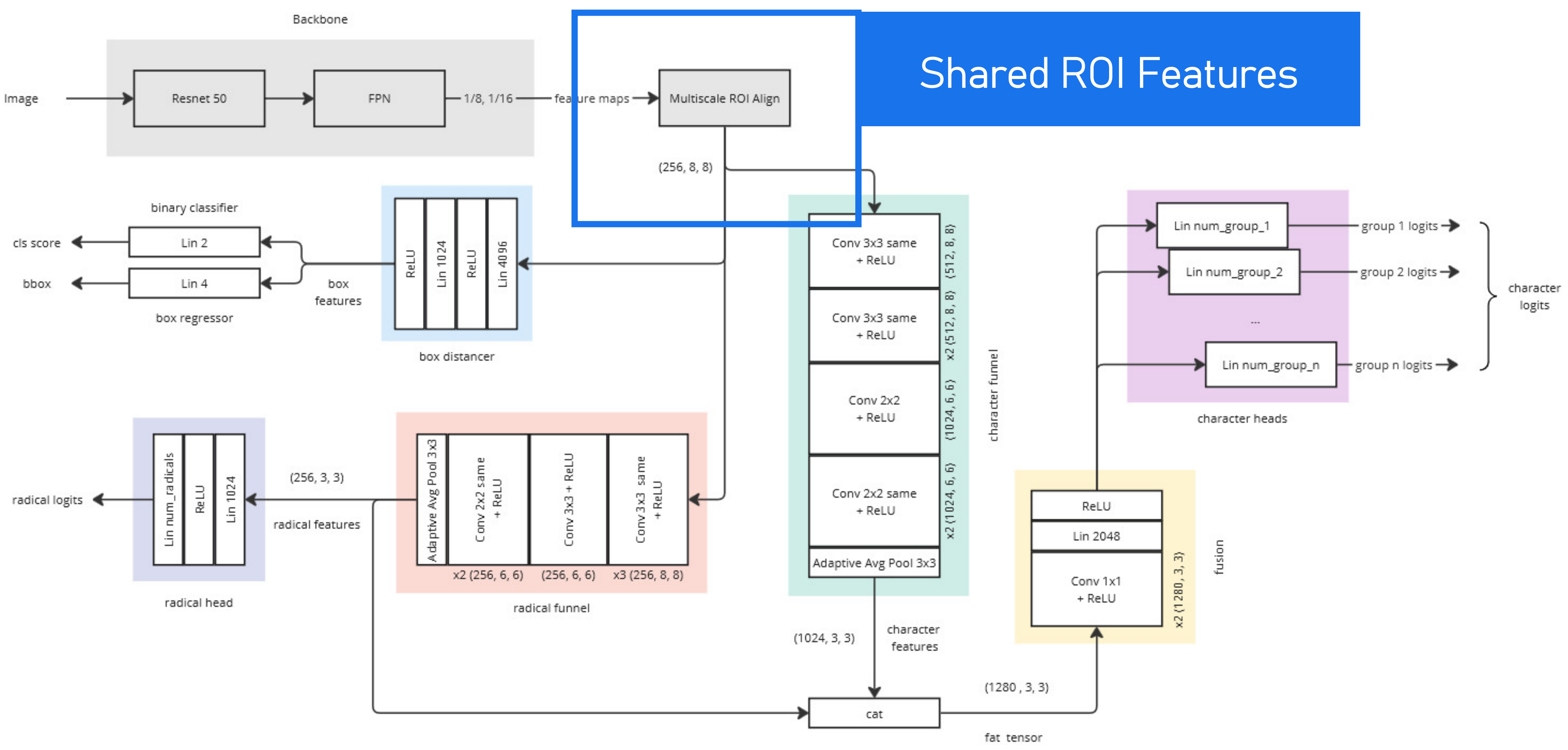
Final Architecture



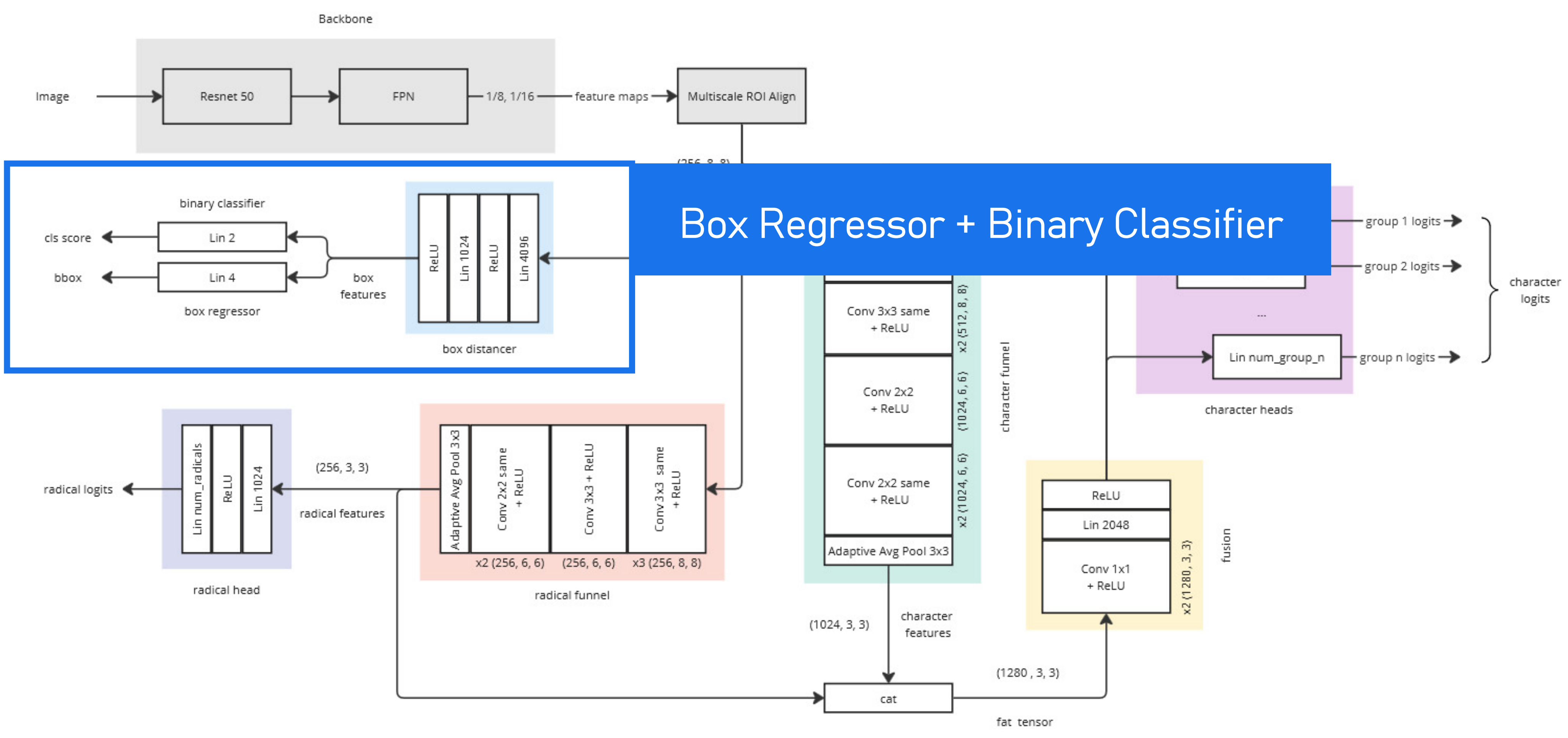
ResNet-50 FPN



Final Architecture



Final Architecture



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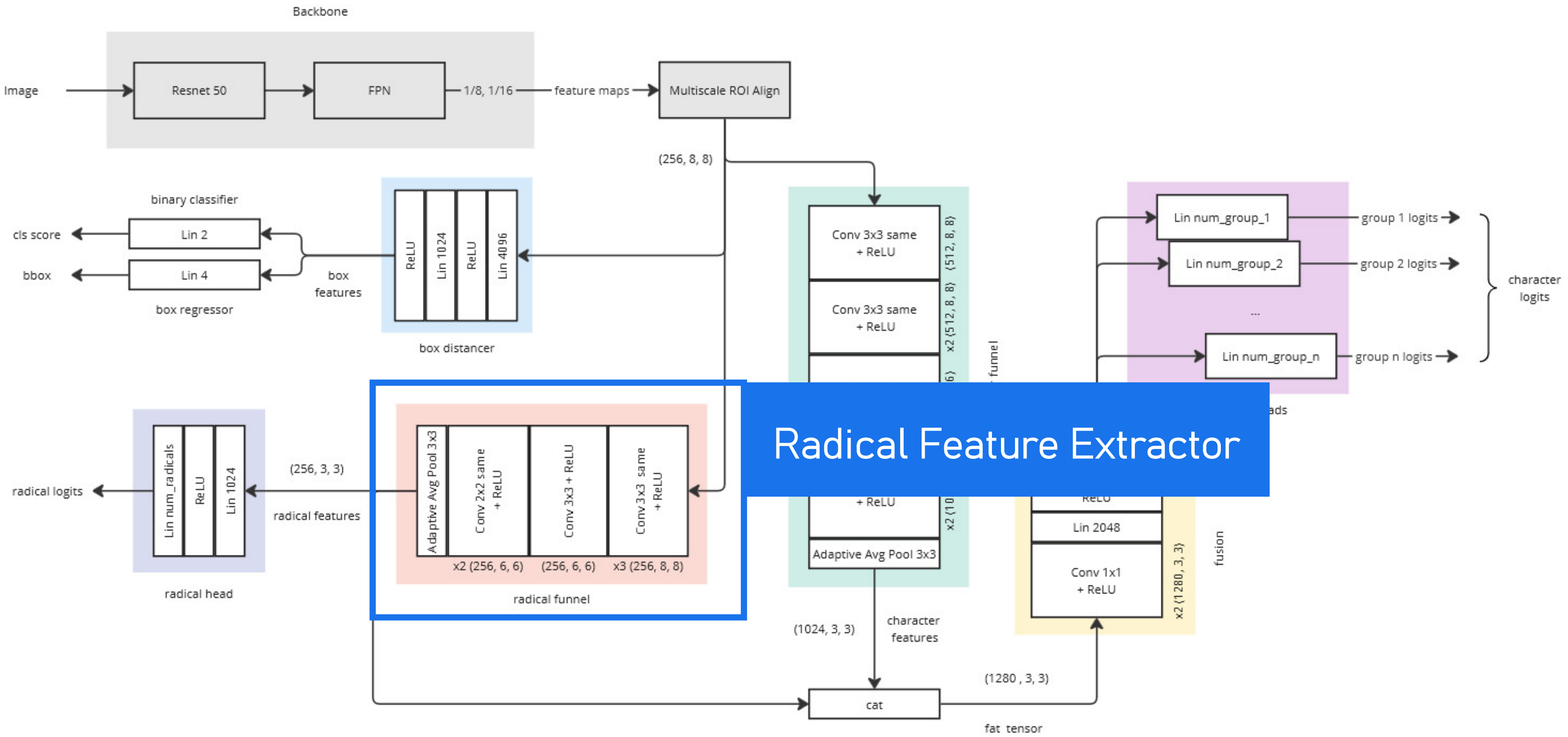

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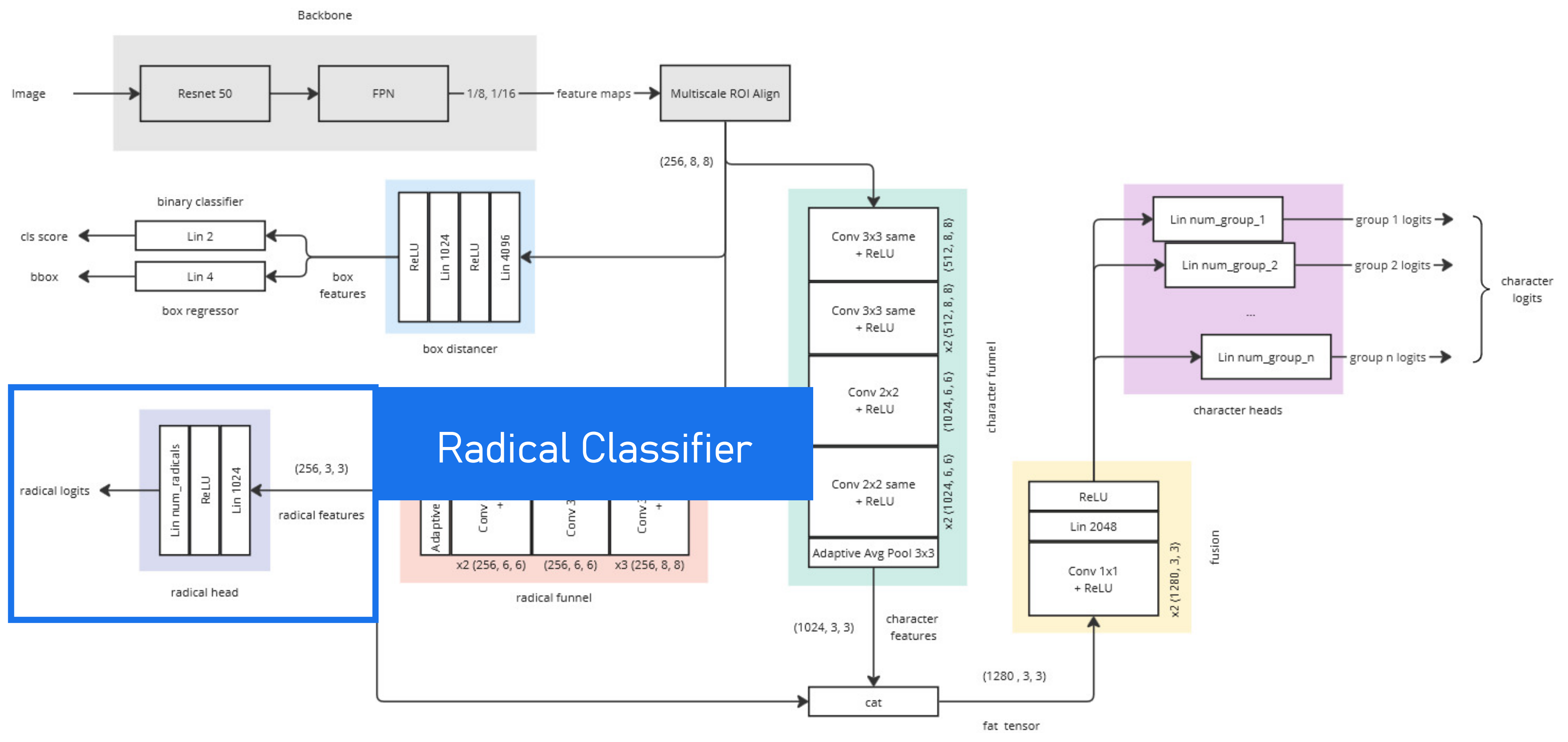
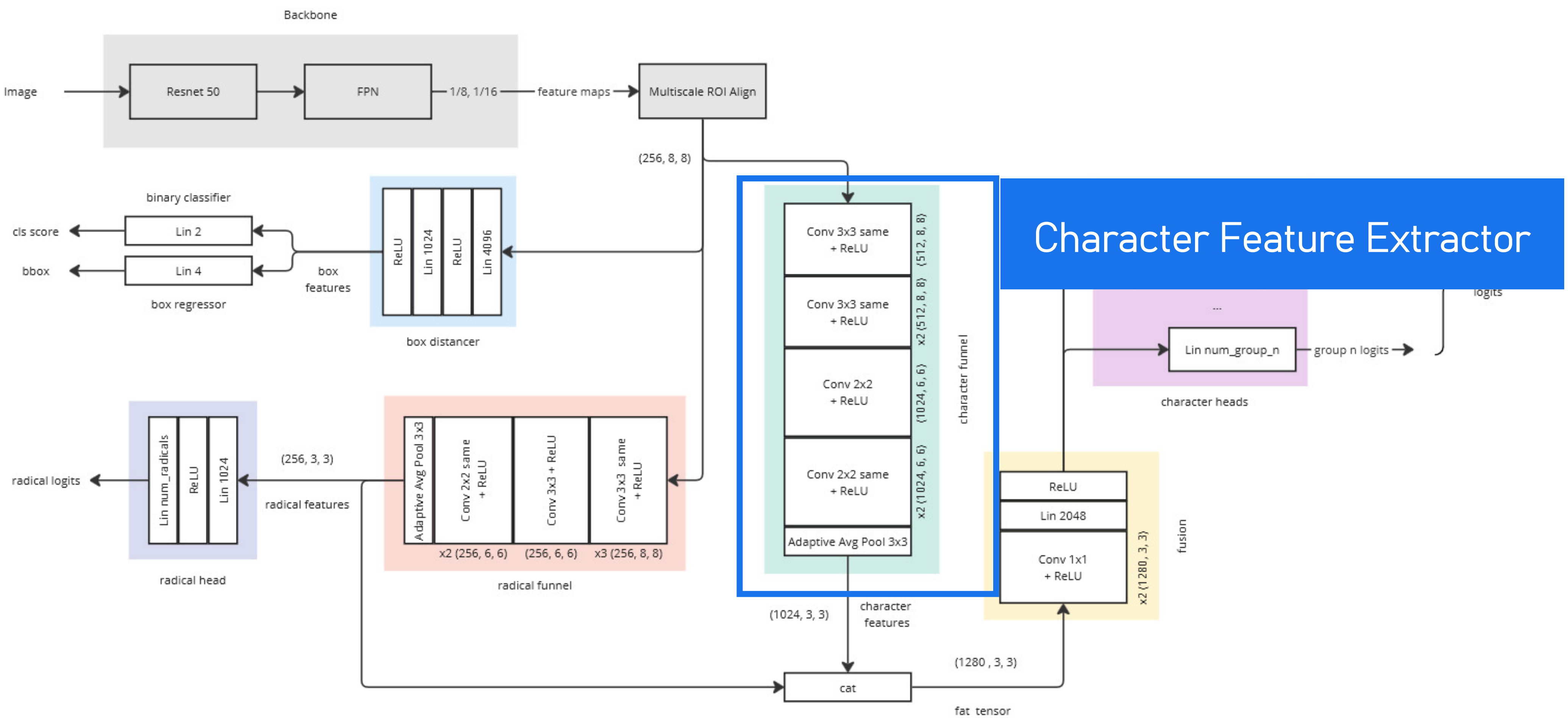
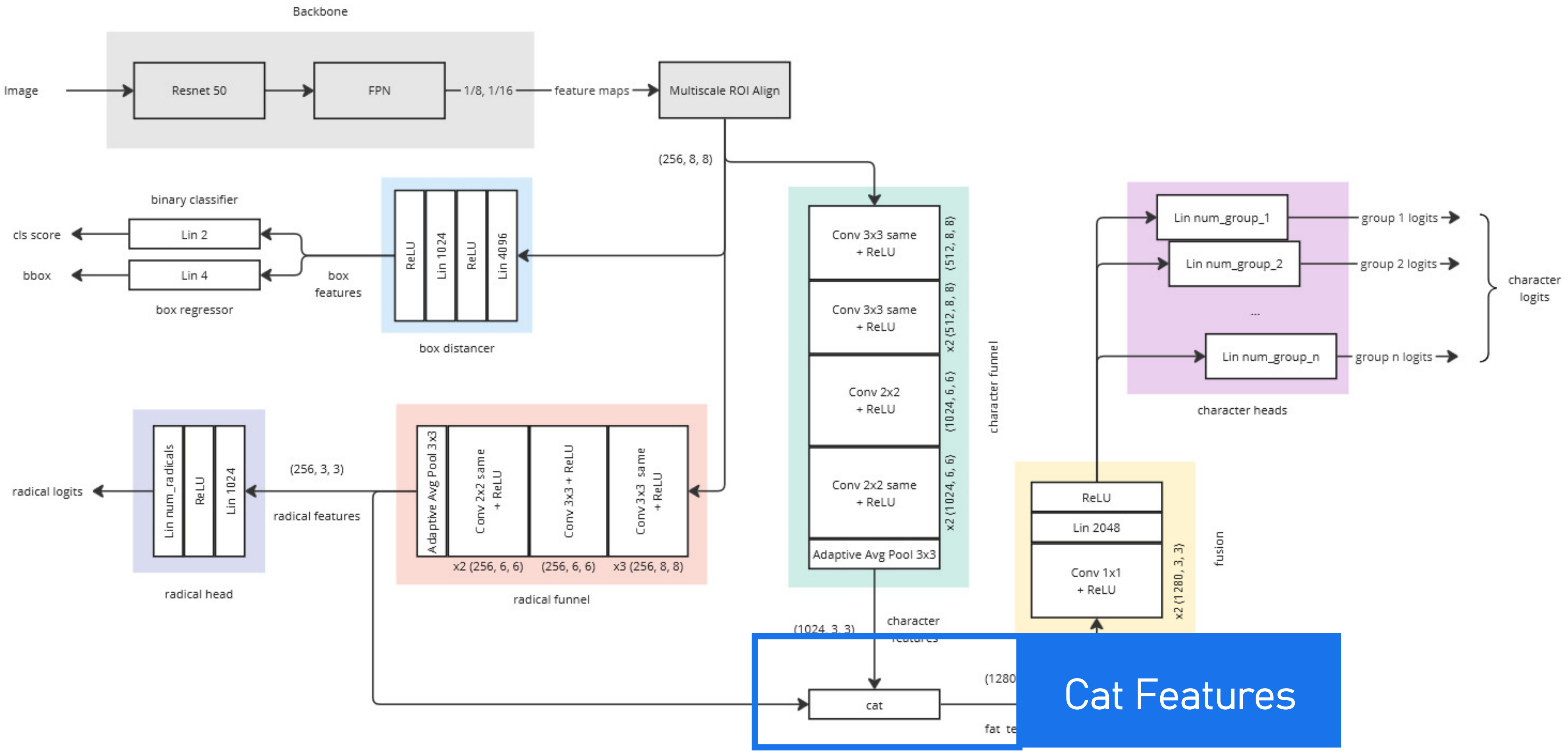


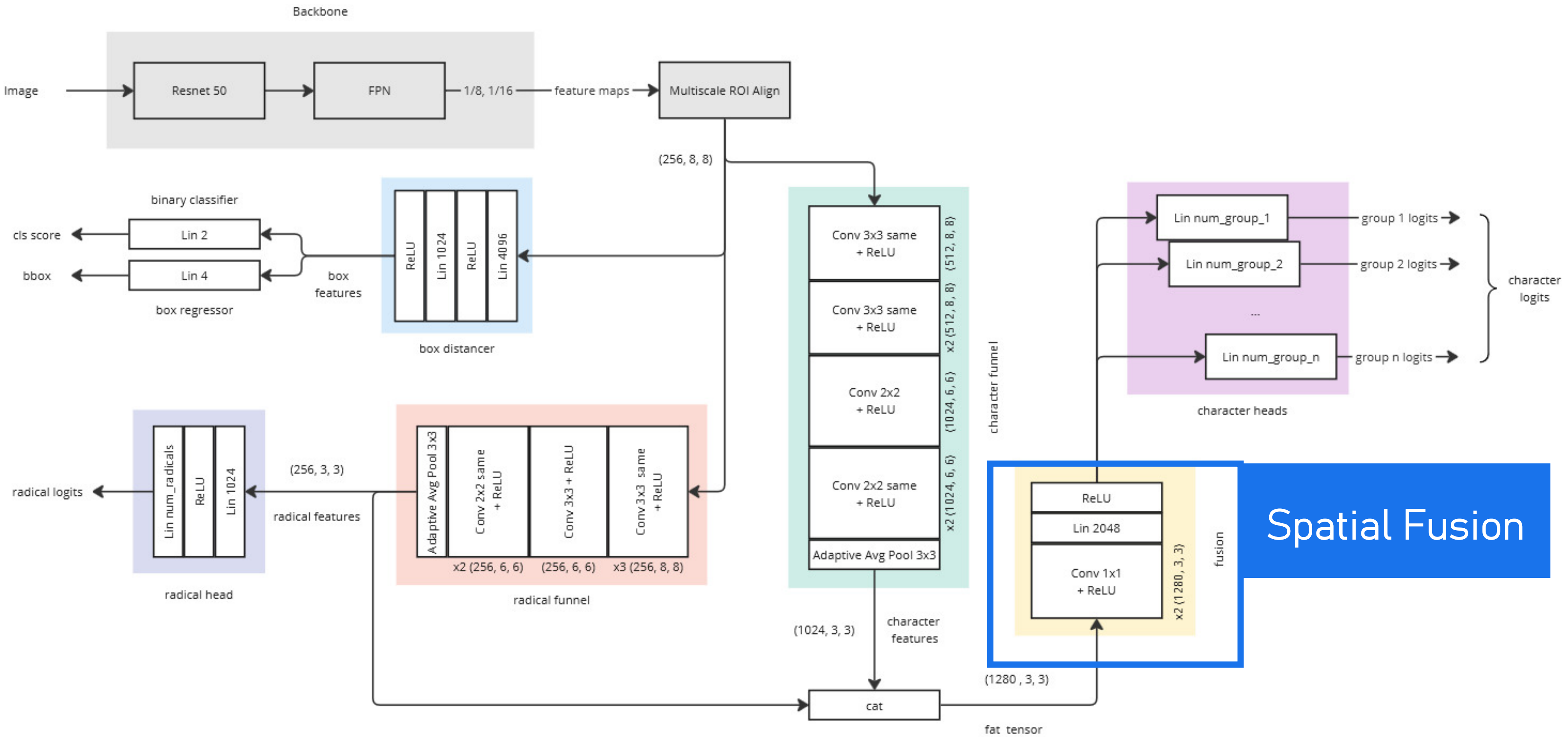
Final Architecture



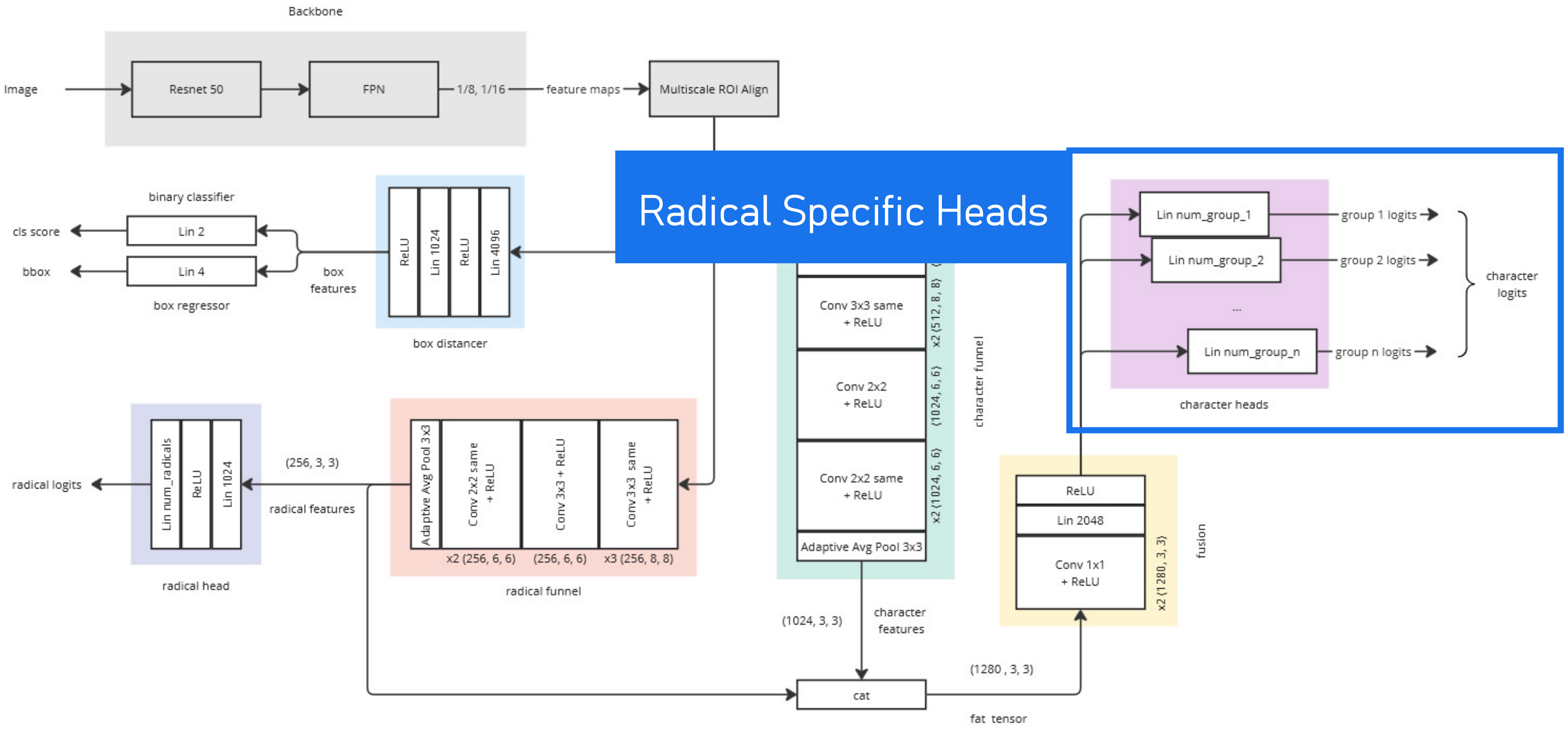
Final Architecture



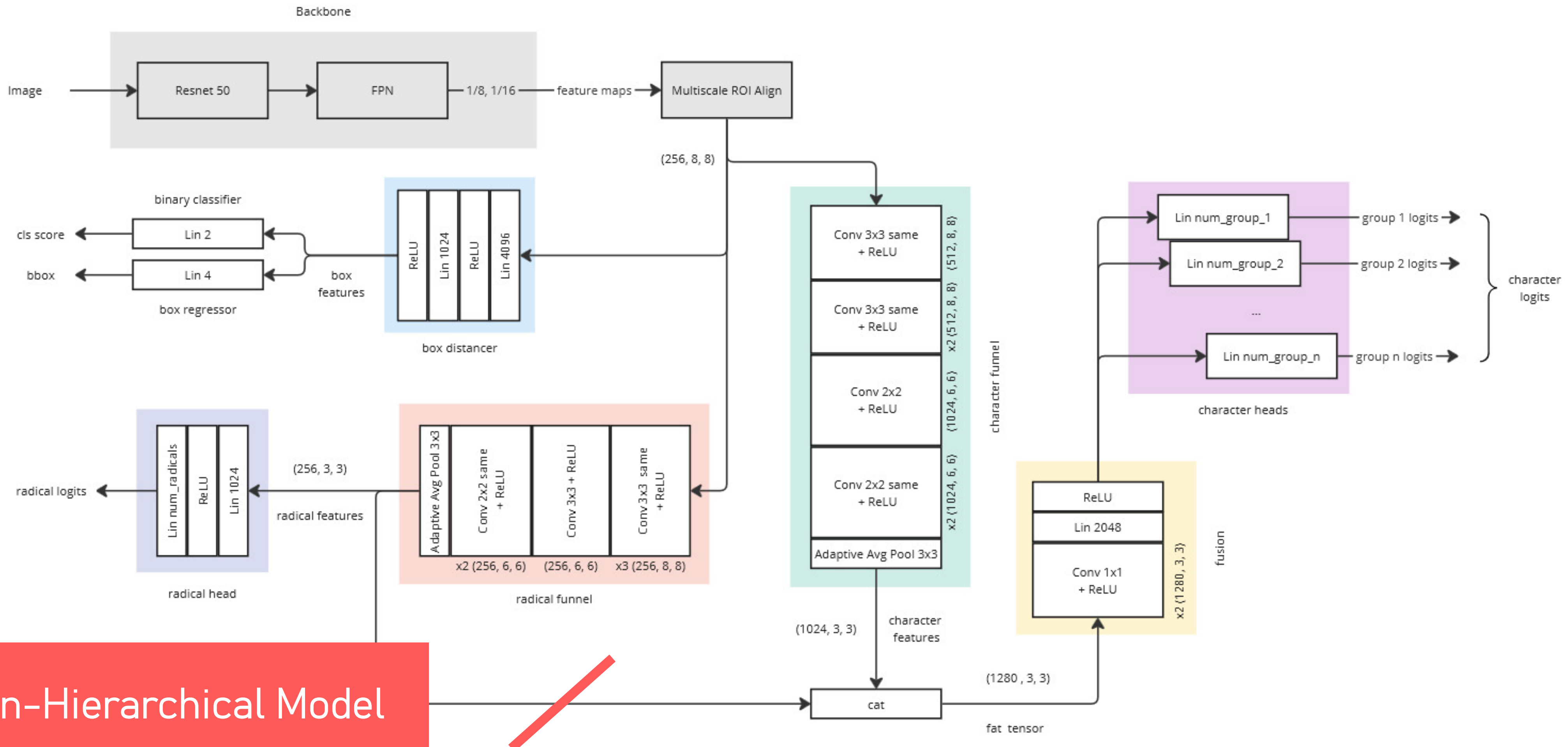
Final Architecture



Final Architecture



Final Architecture



Non-Hierarchical Model

Loss

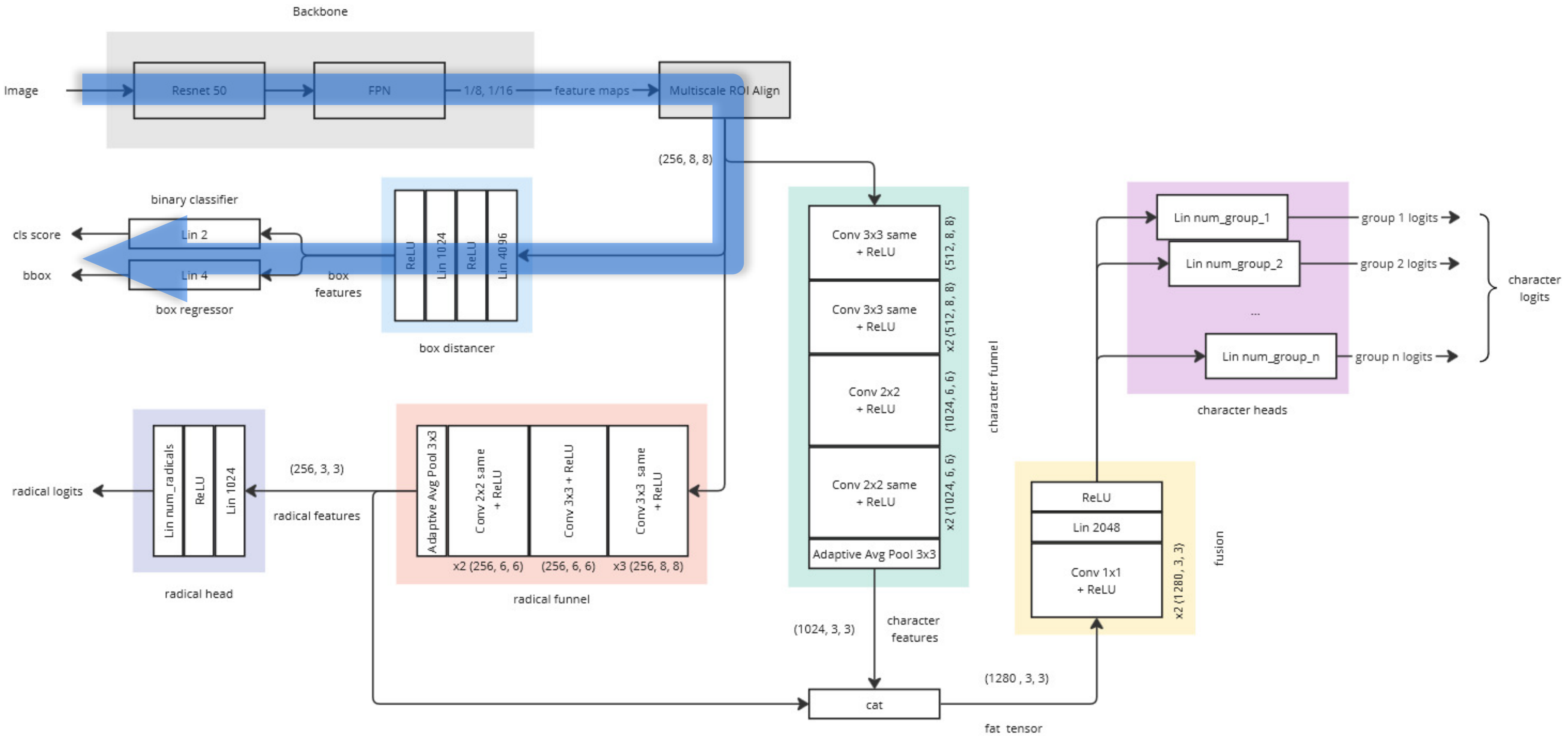
Box Regression
+ Binary Classification Loss

Radical
Classification Loss

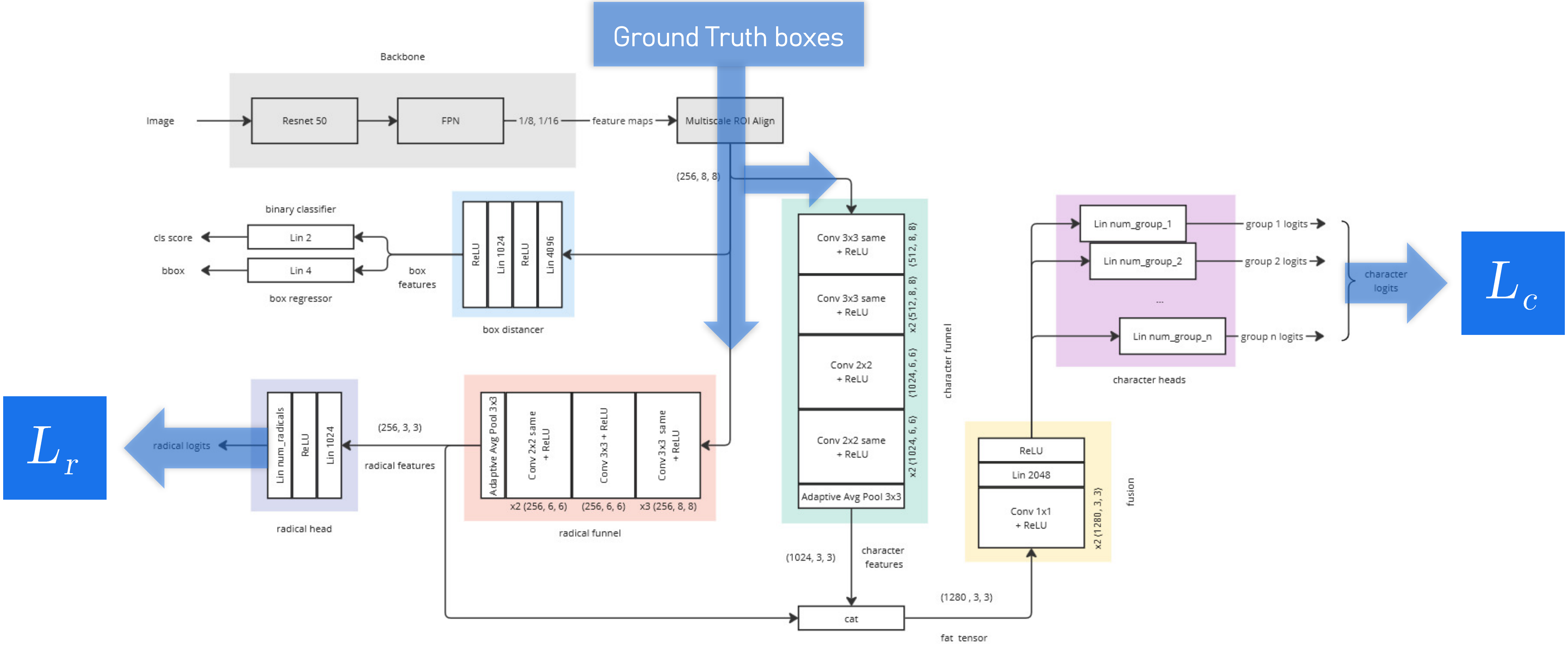
Character
Classification Loss

$$L = \boxed{L_f} + \lambda_r \boxed{L_r} + \lambda_c \boxed{L_c}$$

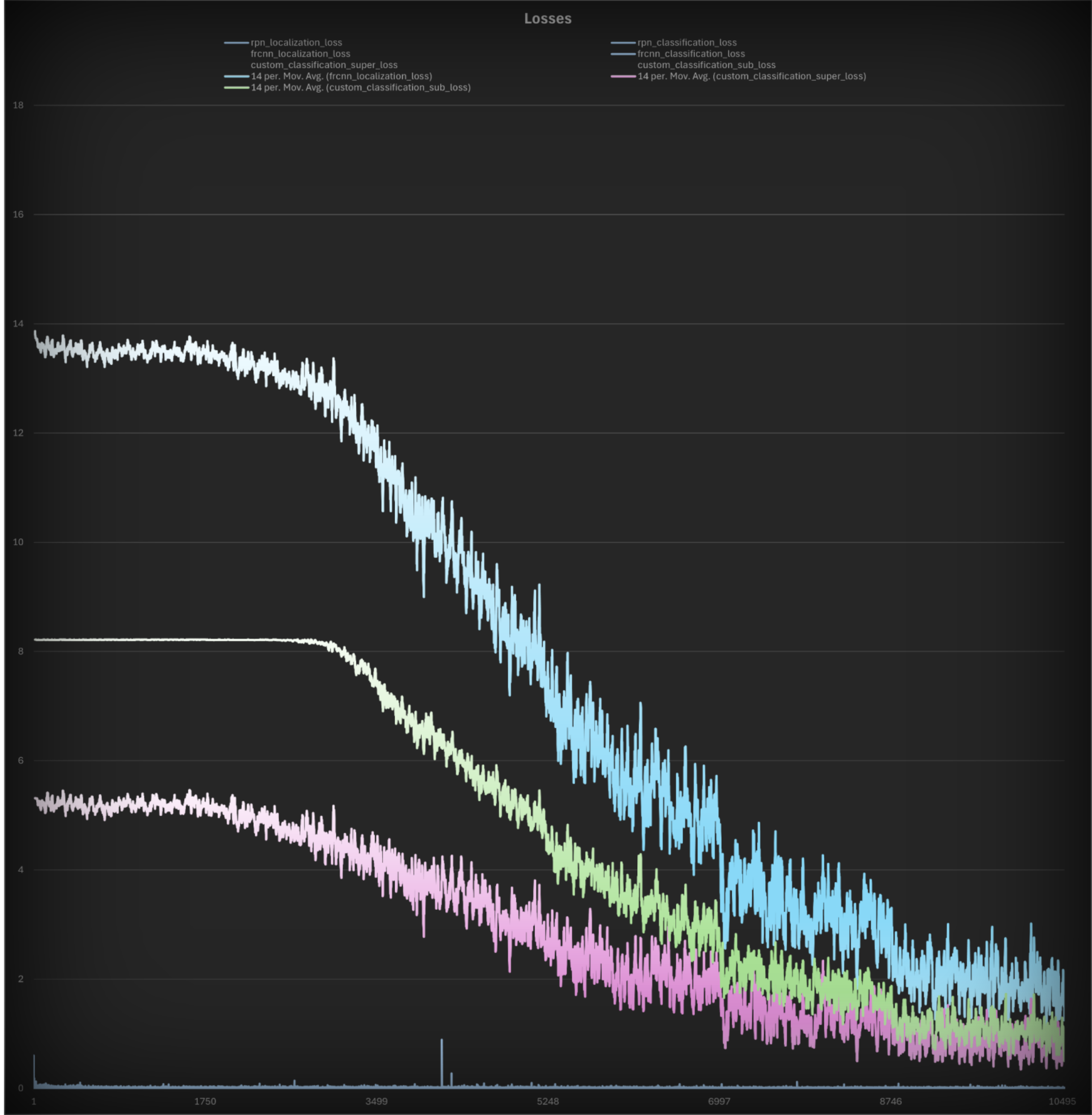
Faster R-CNN Forward Pass



Custom Forward Pass



Experiments



Early experiments

- Fully linear, big latent space dimension (8k), 3GB, ROI size 7
- Provisory 725 and 2k datasets (pruned with Classical Corpus frequency list)
- Showed the need of distancers otherwise gradient/task conflicts with regression loss
- Converged on provisory datasets without Hierarchical Classification
- Only Hierarchical version converged on Full Dataset
- Moved to Convolutional Model: faster, slimmer (145M params, 1GB), more accurate (+10% mAP)
- ROI size increased to 8
- Anchors: {50, 70} with aspect ratios {1:1 , 2:1}

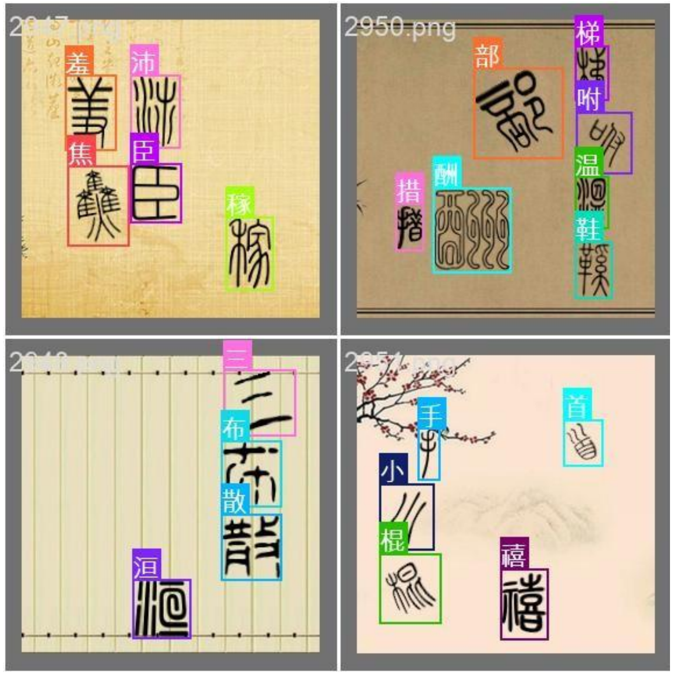
Training

- Adam (per param momentum, fast, less sensitive)? Exploding/unstable loss
- Gradient clipping (max norm 10) after that
- SGD (default values: momentum 0.9, weight decay 1e-5), slow but stable
- Initial Learning Rate 1e-3 + StepLR (gamma 0.8) every epoch (steadiest descent config)
- Batch size 2, Epochs 6
- $\lambda_r = 0.99$, $\lambda_c = 0.95$

Results

YOLO11x (57M, 111MB)

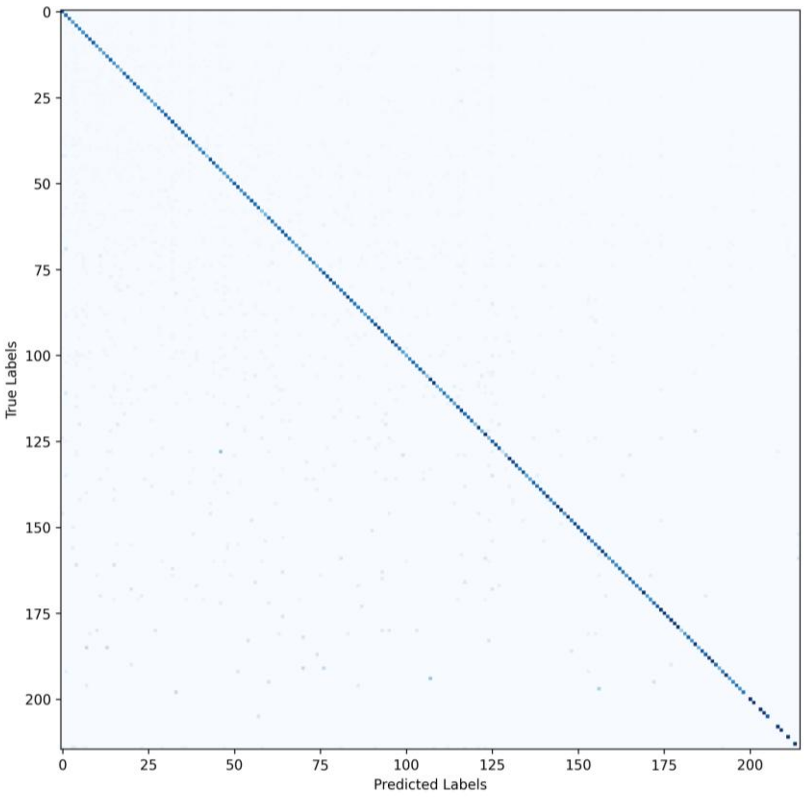
TARGETS



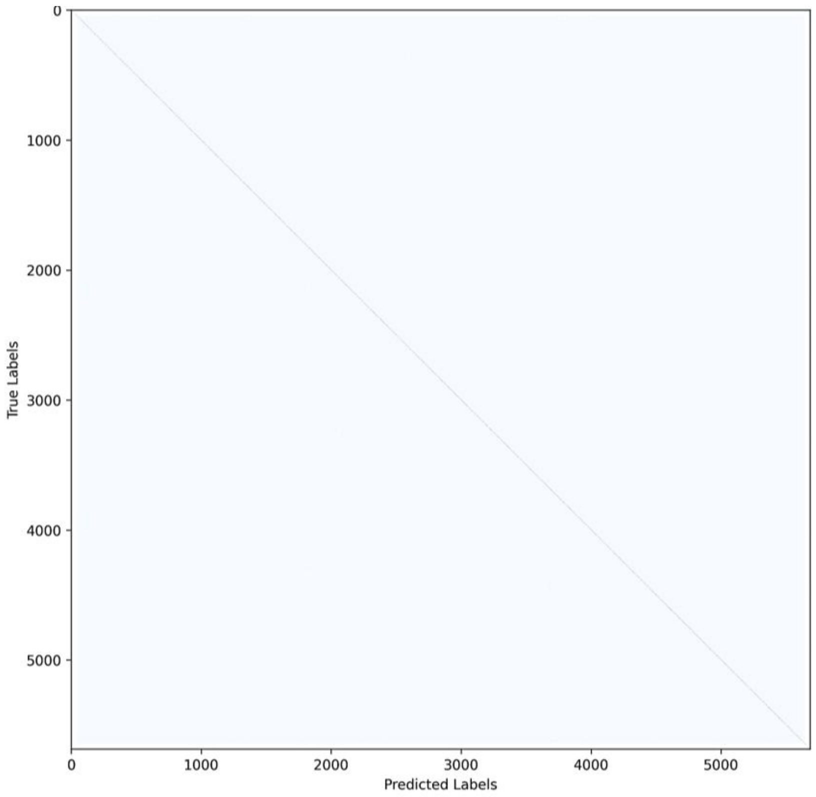
PREDICTIONS

SEAL (Hierarchical)

RADICALS



CHARACTERS

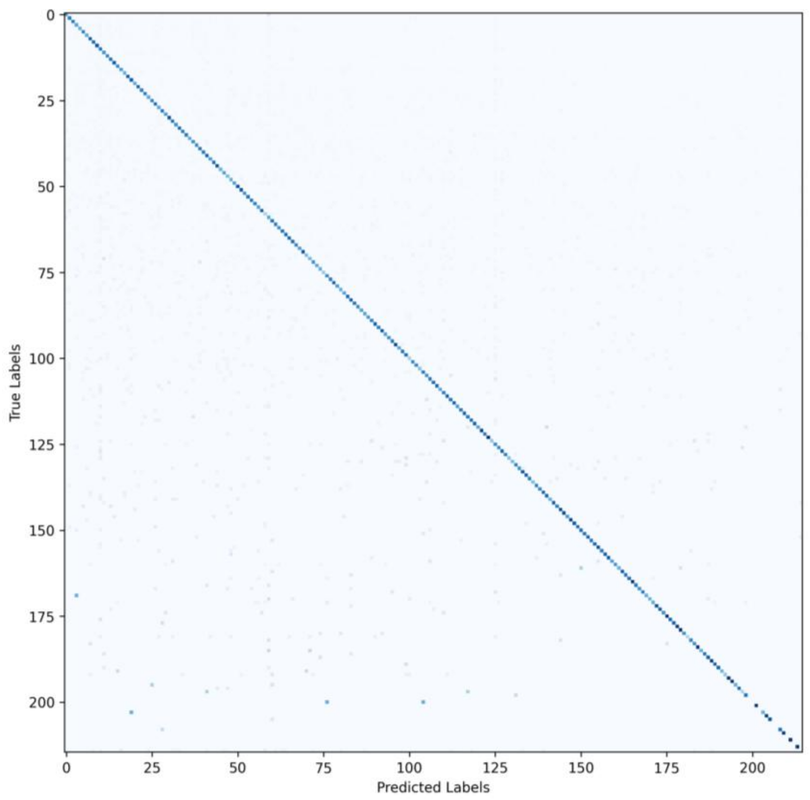


METRICS

mAP@0.75	0.57
mAR	0.65

Confusion Matrices (Non-Hierarchical)

RADICALS



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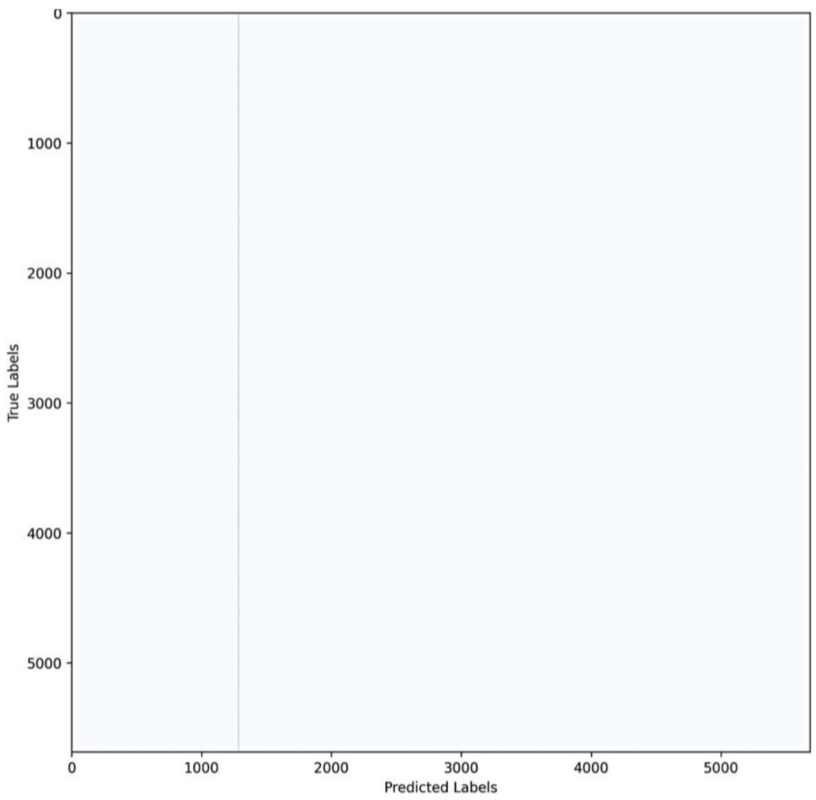
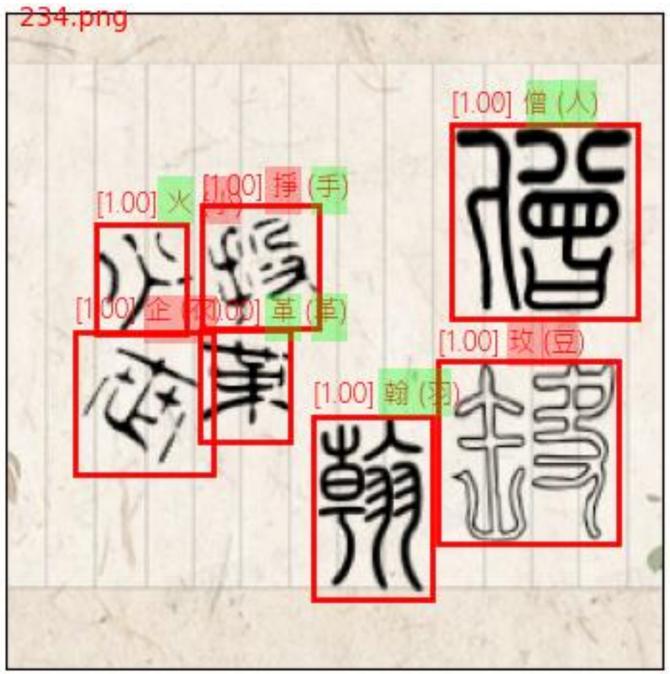


Qualitative inspection



Future Work

- Increase ROI Size
- Increase Backbone Channels
- More dataset augmentations:
    - colors
    - backgrounds
    - deteriorations
    - negative examples
- Junction layer ablation